

Noise or Nuance? Viewing Annotator Disagreement as a Proxy for Subjectivity in Creative Domains

Christine Dierk, Jennifer Healey, Uttaran Bhattacharya, Somdeb Sarkhel

Adobe Research

San Jose, USA

Abstract

Evaluating creative work is challenging for AI due to the absence of any universally agreed-upon set of formal evaluation criteria. While rubrics, frameworks, and expert assessments exist, many of the qualities of creative work are subjective, and human feedback remains critical for seamless AI integration in creative workflows. Subjectivity also naturally leads to disagreements among humans. Traditional annotation approaches often focus on inter-rater agreement and typically collapse disagreements to obtain a single ground-truth label. In this position paper, we argue that in many cases, disagreement is not noise but rather a valuable signal of pluralistic resonances in creative domains. We propose using disagreement as a proxy for subjectivity, discuss how surfacing disagreements can facilitate human-AI alignment, and present strategies for capturing, modeling, and interacting with disagreement.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; *HCI theory, concepts and models*.

Keywords

subjectivity, human-AI collaboration, pluralistic values, creativity

ACM Reference Format:

Christine Dierk, Jennifer Healey, Uttaran Bhattacharya, Somdeb Sarkhel. 2026. Noise or Nuance? Viewing Annotator Disagreement as a Proxy for Subjectivity in Creative Domains. In *Proceedings of CHI Workshop on Bidirectional Human-AI Alignment (CHI '26 BiAlign Workshop)*. ACM, New York, NY, USA, 4 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

As AI assistants and LLM-powered features emerge in creativity support tools, there is a growing need for high-quality human annotations of creative artifacts. Annotators often assess design quality but may also provide judgments on other criteria, such as content type (e.g., commercial vs. editorial photography), graphic design styles (e.g., minimalist, grunge, art, deco), and vibe (e.g., energetic, chill, groovy music). Human annotations of creative artifacts are critical because many of the assessments are at least partially subjective, and there are no formal guidelines or guarantees for what makes a poster eye-catching or a screenplay gut-wrenching. These qualities are not intrinsic to the creative artifact but rather manifest as a resonance between artifact and individual.

Machine learning has traditionally relied on singular ground truth labels to train models and benchmark performance [19, 20] even when this hard coding of labels does not reflect objective

truth [16, 23]. This requirement for single labels has led to the view that, even for subjective tasks, annotator disagreement is merely “noise” to be resolved through majority voting, averaging, or expert adjudication. Such reductionism is known to be harmful for tasks such as hate speech detection where minority opinion, often of the targeted group, can be suppressed as a result [9, 15]. Creative work is inherently subjective, and often lacks a single “right” interpretation. Aggregating diverse annotations into a single label discards nuance [2, 14], as disagreements may be influenced by individual perception and cultural context.

In this position paper, we argue that disagreement can serve as a proxy for subjectivity, capturing multiple distinct but valid interpretations. We advocate leveraging annotator disagreement as useful signal in creative domains and propose that surfacing disagreements can facilitate bidirectional human-AI alignment. Here we present strategies for capturing, modeling, and interacting with disagreement to enable this.

2 Capturing, Modeling, and Interacting with Disagreement

We highlight opportunities to *capture* diverse annotator perspectives, *model* their disagreements, and design *interactions* that make this subjectivity visible and actionable in creative AI systems.

2.0.1 Capturing Disagreement. While annotator disagreement could arise in any annotation task, and often does, annotation methodologies can be designed with subjectivity in mind, explicitly encouraging disagreement [10, 22]. Annotation instructions could be left intentionally vague or ambiguous, encouraging subjective interpretation. Alternatively, annotators could be allowed to select or rank multiple labels [17] rather than identifying a single best match. Regardless of annotation methodology, we advocate for data representations that preserve annotator disagreement over single label aggregations.

Disagreements could be better understood through annotator metadata voluntarily provided or inferred. For example, annotation studies might collect optional demographic or background information from annotators to contextualize differing viewpoints, such as cultural influences [9, 15]. Alternatively, analytical methods could use embeddings to cluster similar perspectives (i.e., similar annotation patterns) into diverse personas. Such personas unlock new opportunities for modeling and interacting with disagreement, mirroring prior work on ensemble and single annotator modeling [8].

2.0.2 Modeling Disagreement. Effective modeling of disagreement requires techniques that preserve individual annotator perspectives. Approaches include predicting soft labels and distributions rather than a single specific label [6, 8, 11, 24], ensemble methods where sub-models are trained on individual annotators [7, 8, 13, 24], and

instruction tuning where explicit guidance and minority perspectives are injected into the training process [24]. As an example, Gordon et al. [12] introduced a supervised learning architecture for a juror model that learns to understand multiple annotator perspectives by explicitly modeling the interaction between content embeddings, annotator identifiers, and sociodemographic attributes. Subsequently, for evaluation, a “jury” can be chosen by sampling from the jurors based on specific compositional goals (e.g., demographic parity or political diversity). We believe that similar results can be achieved for different creative perspectives (e.g. modern, minimalist, maximalist, Wabi-Sabi). Further, modeling disagreements into a distribution of human preferences can lead to richer reward models approximating the underlying subjectivity of preferences, and, in turn lead to pluralistically-aligned generative models [25]. Regardless of system architecture, we argue that creative AI systems should explicitly represent uncertainty and subjectivity, supporting pluralistic interpretations.

2.0.3 Interacting with Disagreement. Surfacing model uncertainty and diverse perspective could enable new forms of human-AI collaboration. If annotator disagreements can be modeled by synthetic personas, AI systems can be trained to reflect the distinct viewpoints these personas represent. An ensemble of such models would allow for multiple perspectives to give a nuanced interpretation of a creative work [5]. Furthermore, when collaborating with AI, the system could be transparent about the ensemble or persona it’s modeling. This would allow AI to highlight or surface nuanced recommendations with options instead of making a singular recommendation. When giving feedback on a poster for example, the AI could give a diversity of opinion based on various audiences: “a child might find this poster overly scary, you should consider eliminating ...” or “this type of representation may have the following unintended interpretation ...” or even “this work may be seen as controversial because...”. This informed surfacing of diverse perspectives can allow humans to think more critically about the AI’s recommendations. For example, if the creator was intentionally trying to attract an adult audience to a horror film, they may choose to ignore or override the AI’s suggestion regarding child-friendliness. Conversely, if a diverse perspective AI could look at a creation from multiple viewpoints [1], it might surface how the verbiage or imagery could be interpreted by others that are potentially outside of the user’s cultural understanding [3, 4, 18].

3 Disagreement as a Proxy for Subjectivity

Annotator disagreement can be used as a proxy for task subjectivity within a single domain. Tasks with consistently high disagreement are likely more subjective and open to creative interpretation. Alternatively, tasks where annotators converge are likely to be more clear cut exemplars. For example, a graphic design with low contrast and illegible text may be universally interpreted as bad design, indicating low subjectivity; whereas a bold maximalist design may be rated highly by one annotator who finds it eye-catching, and rated lower by another who finds it cluttered and distracting.

In evaluation, disagreement as a measure of task subjectivity enables informed segmentation of benchmark data. Evaluators can identify highly subjective and ambiguous tasks, either focusing on them or excluding them from analysis and alignment efforts

depending on research goals. For example, an evaluator may enforce greater alignment standards for clear cut exemplars with low annotator disagreement while relaxing requirements for more subjective tasks. Alternatively, another evaluator may focus prompt engineering efforts on improving alignment for a subset of highly subjective tasks with high annotator disagreement.

In modeling, the ability to predict annotator disagreement and thus estimate a task’s underlying subjectivity opens additional opportunities for system design. For highly subjective tasks, the system might proactively generate multiple candidate outputs to reflect the range of possible interpretations. A system might also leverage high-disagreement tasks to elicit user preferences. Understanding how a user perceives a highly subjective stimulus might indicate which interpretations and personas they most align with.

4 Towards Bidirectional Human-AI Alignment

Using the bidirectional human-AI alignment framework proposed by Shen et al. [21], we describe how preserving annotator disagreement can (1) allow AI systems to more closely align with human perception, and (2) enable humans to more effectively understand and collaborate with AI systems.

4.0.1 Aligning AI with Humans. Perfect agreement rarely occurs in human annotation, especially for subjective tasks. Why should creative AI be limited to single label predictions? Pluralistic perspectives may more closely approximate how humans perceive creative work—not as fixed mappings between content and labels, but as a diverse space of potential interpretations informed by taste, background, and cultural context. An AI system that can predict expected disagreement and use it as a proxy for subjectivity may also be more human-aligned. Such a system might express greater uncertainty when providing suggestions for what it identifies as highly subjective tasks. Or, it could present multiple interpretations over concrete guidance. Finally, annotator-derived personas may be more aligned than a single interpretation based on aggregate scores. Personalization is possible when the AI system can predict which persona most closely matches the goals of the user.

4.0.2 Aligning Humans with AI. Expressing uncertainty and presenting pluralistic interpretations may make an AI system more transparent and interpretable, making it easier for humans to understand and collaborate with AI. Encountering diverse perspectives might encourage humans to think more critically about AI recommendations. Users may even begin to view AI as a tool for reflection instead of an authoritative expert. Annotator-driven personas can support human agency as users simulate audience reception, identify unintended interpretations, and decide which personas best align with their creative goals.

5 Conclusion

In this position paper, we argued that in many cases annotator disagreement is not merely noise but a valuable signal for understanding subjectivity in creative domains. We suggested strategies for capturing, modeling, and interacting with disagreement, and discussed how preserving annotator disagreement could support bidirectional human-AI alignment.

References

- [1] History of Architecture Architectural Reviews. 2018. The Controversy of the Louvre Pyramid. <https://draftingharmony.wordpress.com/2025/01/23/the-controversy-of-the-louvre-pyramid/>.
- [2] Joris Baan, Wilker Aziz, Barbara Plank, and Raquel Fernandez. 2022. Stop Measuring Calibration When Humans Disagree. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang (Eds.). Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 1892–1915. doi:10.18653/v1/2022.emnlp-main.124
- [3] BBC. 2018. H&M apologises over 'racist' image of black boy in hoodie. <https://www.bbc.com/news/newsbeat-42603960>.
- [4] Sesali Bowen. 2017. I Tried Watching Guerrilla — & As A Black Woman, I Just Can't. <https://www.refinery29.com/en-us/2017/04/150292/guerrilla-showtime-black-power-movement-misogyny-sexism>.
- [5] Yoonseo Choi, Eun Jeong Kang, Seulgi Choi, Min Kyung Lee, and Juho Kim. 2025. Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 149, 32 pages. doi:10.1145/3706598.3714034
- [6] Huang-Cheng Chou and Chi-Chun Lee. 2019. Every Rating Matters: Joint Learning of Subjective Labels and Individual Annotators for Speech Emotion Classification. In *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 5886–5890. doi:10.1109/ICASSP.2019.8682170
- [7] Aida Mostafazadeh Davani, Mark Diaz, and Vinodkumar Prabhakaran. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics* 10 (01 2022), 92–110. doi:10.1162/tacl_a_00449
- [8] H.M. Fayek, M. Lech, and L. Cavedon. 2016. Modeling subjectiveness in emotion recognition with deep neural networks: Ensembles vs soft labels. In *2016 International Joint Conference on Neural Networks (IJCNN)*. 566–570. doi:10.1109/IJCNN.2016.7727250
- [9] Eve Fleisig, Rediet Abebe, and Dan Klein. 2023. When the Majority is Wrong: Modeling Annotator Disagreement for Subjective Tasks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 6715–6726. doi:10.18653/v1/2023.emnlp-main.415
- [10] Tommaso Fornaciari, Alexandra Uma, Silviu Paun, Barbara Plank, Dirk Hovy, and Massimo Poesio. 2021. Beyond Black & White: Leveraging Annotator Disagreement via Soft-Label Multi-Task Learning. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou (Eds.). Association for Computational Linguistics, Online, 2591–2597. doi:10.18653/v1/2021.naacl-main.204
- [11] Tommaso Fornaciari, Alexandra Uma, Silviu Paun, Barbara Plank, Dirk Hovy, Massimo Poesio, et al. 2021. Beyond black & white: Leveraging annotator disagreement via soft-label multi-task learning. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics.
- [12] Mitchell L. Gordon, Michelle S. Lam, Joon Sung Park, Kayur Patel, Jeff Hancock, Tatsunori Hashimoto, and Michael S. Bernstein. 2022. Jury Learning: Integrating Dissenting Voices into Machine Learning Models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (New Orleans, LA, USA) (CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 115, 19 pages. doi:10.1145/3491102.3502004
- [13] Melody Guan, Varun Gulshan, Andrew Dai, and Geoffrey Hinton. 2018. Who said what: Modeling individual labelers improves classification. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 32.
- [14] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. 2025. Artificial Hive-mind: The Open-Ended Homogeneity of Language Models (and Beyond). In *Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (San Diego, CA) (NeurIPS '25)*. <https://neurips.cc/virtual/2025/loc/san-diego/poster/121421>
- [15] Savannah Larimore, Ian Kennedy, Breon Haskett, and Alina Arseniev-Koehler. 2021. Reconsidering Annotator Disagreement about Racist Language: Noise or Signal?. In *Proceedings of the Ninth International Workshop on Natural Language Processing for Social Media*, David Jurgens, Saif M. Mohammad, Stefan Proisl, and Josef Ruppenhofer (Eds.). Association for Computational Linguistics, Online, 81–90. <https://aclanthology.org/2021.socialnlp-1.7>
- [16] UCI Machine Learning. 2016. Mushroom Classification. <https://www.kaggle.com/datasets/uciml/mushroom-classification>.
- [17] Marian Marchal, Merel Scholman, Frances Yung, and Vera Demberg. 2022. Establishing Annotation Quality in Multi-label Annotations. In *Proceedings of the 29th International Conference on Computational Linguistics*, Nicoletta Calzolari, Churen Huang, Hansaem Kim, James Pustejovsky, Leo Wanner, Key-Sun Choi, Pum-Mo Ryu, Hsin-Hsi Chen, Lucia Donatelli, Heng Ji, Sadao Kurohashi, Patrizia Pagio, Nianwen Xue, Seokhwan Kim, Younggyun Hahm, Zhong He, Tony Kyungil Lee, Enrico Santus, Francis Bond, and Seung-Hoon Na (Eds.). International Committee on Computational Linguistics, Gyeongju, Republic of Korea, 3659–3668. <https://aclanthology.org/2022.coling-1.322/>
- [18] Scott Mautz. 2020. A Peloton Ad Sparked Huge Controversy Over Its Sexism. It's Also Just a Terrible Commercial. <https://scottmautz.com/a-peloton-ad-sparked-huge-controversy-over-its-sexism-its-also-just-a-terrible-commercial/>.
- [19] Barbara Plank. 2022. The Problem of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. *arXiv preprint arXiv:2211.02570* (2022).
- [20] Barbara Plank, Dirk Hovy, and Anders Søgaard. 2014. Learning part-of-speech taggers with inter-annotator agreement loss. In *Proceedings of the 14th Conference of the European Chapter of the Association for Computational Linguistics*, Shuly Wintner, Sharon Goldwater, and Stefan Riezler (Eds.). Association for Computational Linguistics, Gothenburg, Sweden, 742–751. doi:10.3115/v1/E14-1078
- [21] Hua Shen, Tiffany Kneare, Reshmi Ghosh, Kenan Alkiek, Kundan Krishna, Yachuan Liu, Ziqiao Ma, Savvas Petridis, Yi-Hao Peng, Li Qiwei, et al. 2024. Towards bidirectional human-ai alignment: A systematic review for clarifications, framework, and future directions. *arXiv preprint arXiv:2406.09264* 2406 (2024), 1–56.
- [22] Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2022. Learning from Disagreement: A Survey. *J. Artif. Int. Res.* 72 (Jan. 2022), 1385–1470. doi:10.1613/jair.1.12752
- [23] Kiri L. Wagstaff. 2012. Machine learning that matters. In *Proceedings of the 29th International Conference on International Conference on Machine Learning (Edinburgh, Scotland) (ICML '12)*. Omnipress, Madison, WI, USA, 1851–1856.
- [24] Jin Xu, Mariët Theune, and Daniel Braun. 2024. Leveraging annotator disagreement for text classification. *arXiv preprint arXiv:2409.17577* (2024).
- [25] Michael Jq Zhang, Zhilin Wang, Jena D. Hwang, Yi Dong, Olivier Delalleau, Yejin Choi, Eunsol Choi, Xiang Ren, and Valentina Pyatkin. 2025. Diverging Preferences: When do Annotators Disagree and do Models Know?. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*, Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu (Eds.). PMLR, 76193–76212. <https://proceedings.mlr.press/v267/zhang25bx.html>